Ans to the Question NO - 01

Demand:

Demand means desire to have a commodity with ablity and willingness to pay. There are three componants of demand

1. Desire
2. ability to pay
3. Willingness to pay

low of demand: low of demand is the inverse relationship between price and demand. If price increase demand decrease and If price decrease the demand increase. It is called the low of demand,

Formula of low of demand:

$$D_m = f(P_m, Y, P_o, T, A, E_f, N)$$

Demand Function: when we express the relationship between demand and its determinant mathematically, The relationship is know as demand function.

Ans to the Question No-02

Price Elasticity of demand: The price elasticity of demand ~~te~~ is defined as the responsiveness of demand due to change in price.

Given,

Monthly consumer income = 50000

Income of consumer increase 10%

$$= 50000 \times \frac{110}{100}$$

$$= 55000\cancel{0}$$

Relative change in Income $\frac{\Delta Y}{Y} = \frac{55000\cancel{0} - 50000}{50000}$

$$= \frac{1}{10}$$

Relative change in demand $= \frac{\Delta Q}{Q} = \frac{14-12}{12} = \frac{1}{6}$

$\therefore$ Elasticity of demand $E_y = \dfrac{\frac{\Delta Q}{Q}}{\frac{\Delta Y}{Y}} = \dfrac{\frac{1}{6}}{\frac{1}{10}} = 1.6667.$

$\therefore$ $E_y > 1$ ~~Highly Elasted~~ Superior commodity.

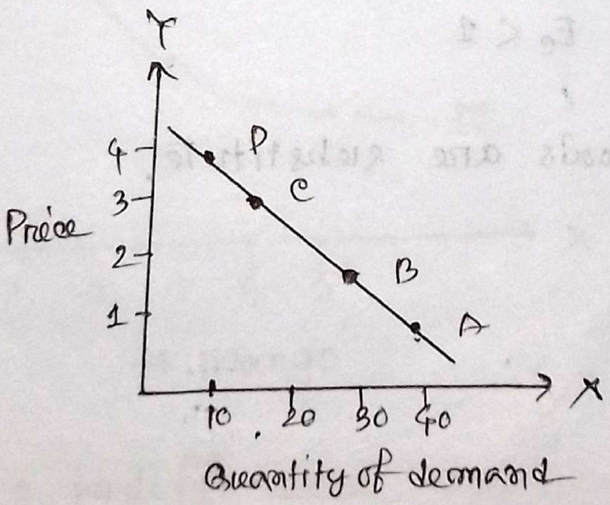
Ans to the Question No - 03.

Demand curve: The graphical representation of demand schedule is called demand curve.

Demand schedule: The tabular representation of price and demand is called demand schedule.

Price	demand	point
1	40	A
2	30	B
3	20	C
4	10	D

demand schedule



Ans to the Question No – 04

Cross Elasticity of demand: Cross elasticity of demand defined as the responsiveness of demand due to change in related goods price.

Solve:

$$\text{Relative change in demand} = \frac{\Delta Q_x}{Q_x} = \frac{5}{100} = \frac{1}{20}$$

$$\text{Relative change in price of other commodity} = \frac{\Delta P_y}{P_y} = \frac{55-50}{50} = \frac{5}{50} = \frac{1}{10}$$

$$\therefore \text{Elasticity of demand } E_c = \frac{\frac{\Delta Q_x}{Q_x}}{\frac{\Delta P_y}{P_y}} = \frac{\frac{1}{20}}{\frac{1}{10}} = \frac{1}{20} \times \frac{10}{1} = \frac{1}{2}$$

$$E_c < 1$$

$\therefore$ Goods are substitute.

Tabular representation of indifference curve

Point	X commodity	Y commodity	MRS
A	1	15	—
B	2	11	1 : 4
C	3	8	A : 3
D	4	6	1 : 2
E	5	5	1 : 1

Graphical representation of indifference curve

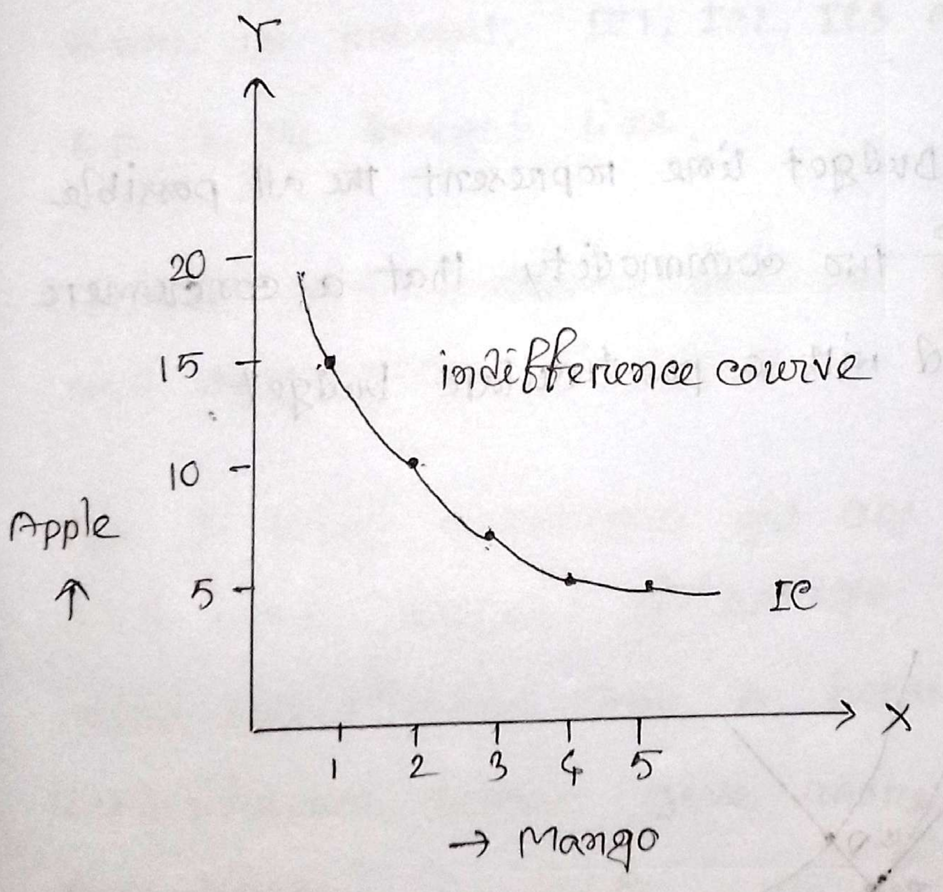


Fig : indifference curve

Ans to the Question No – 5

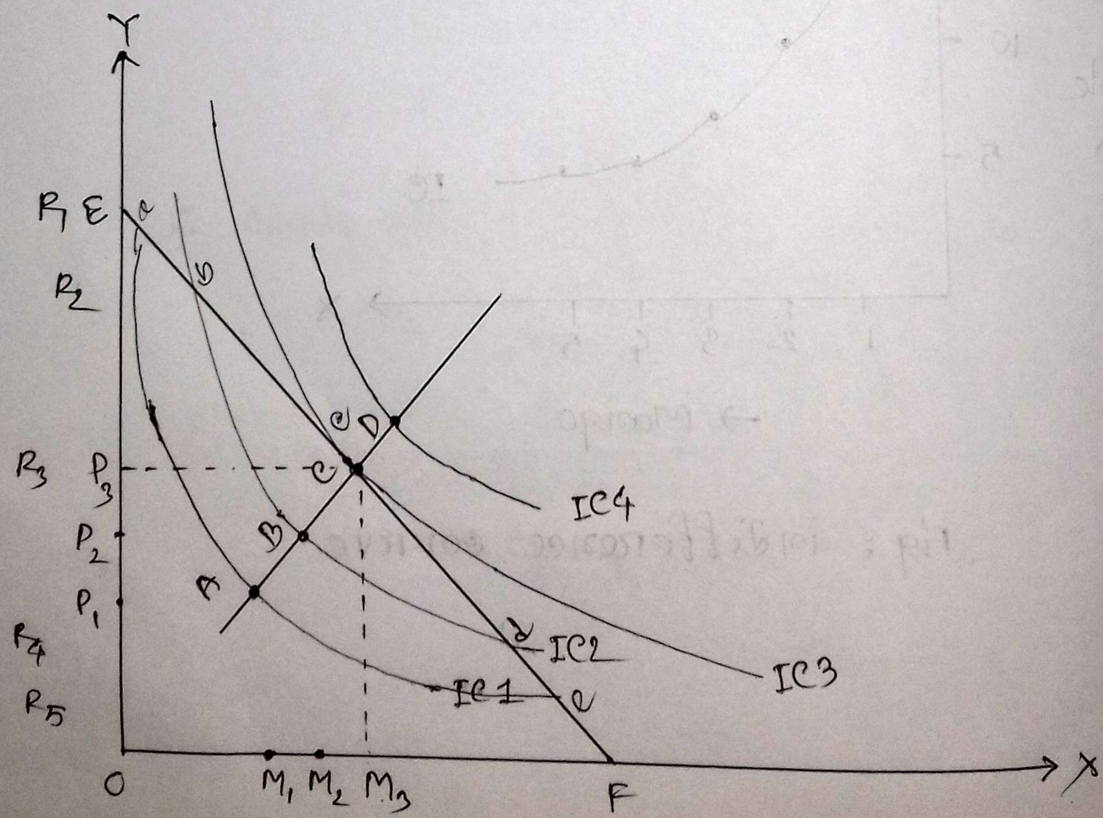
Consumer's equilibrium:

For showing consumer equilibrium we require

1. Indifference curve.
2. Budget line.

Indifference curve: Indifference curve is the curve that represt the combination of two or more commodity that give the same satisfaction to the consumer.

Budget line: Budget line represent the all possible combination of two commodity that a consumer can be purchased with a particular budget.



Consumer equilibrium:

Equilibrium is the state of balance which no deviation is desired. consumer equilibrium is the point which consumer get maximum satisfaction with his/her budget.

Discussion: In this figure there are four indifference courve are present, IC1, IC2, IC3 and IC4 and EF is the Budget line.

For A poin consumer get OR1 amount of apple and OM1 amout of Mange.

For B point consumer get OR2 amount of apple and OM2 amount of mango, this point give more statisfaction than A. because upper indifference courve give more satisfaction than lower. ∴ B > A.

Similarly poin (C > B) give more satisfaction than B.